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EDUCATION

15/02/2021 – 25/06/2025

Ph.D., Biomedical Sciences & Electrical Engineering

KU Leuven, Leuven, Belgium (award date: 25/06/2025)

Ph.D thesis: From Sound to Meaning: Modeling Speech Perception in the Brain with Artificial Neural Networks

01/02/2018 – 15/03/2020

M.Sc., Bioengineering

EPFL, Lausanne, Switzerland

Master thesis: Effects of Epidural Electrical Stimulation on the ascending proprioceptive circuits measured intracranially in Rhesus monkeys.

01/09/2013 – 15/12/2017

B.Sc., Life Science Engineering

EPFL, Lausanne, Switzerland

ACADEMIC EXPERIENCE

25/06/2025 – present

Postdoctoral Researcher, Dept. of Neuroscience & Electrical Engineering,
KU Leuven, ExpORL/PSI, Belgium

14/03/2026 – 15/07/2026	Visiting Scholar , Stanford University, Gwilliams Laboratory of Speech Neuroscience, USA <i>Project: Single neuron encoding of acoustic and linguistic information in the human hippocampus during story listening</i> Supervisor: Prof. Laura Gwilliams
14/06/2024 – 15/10/2024	Visiting Scholar , Columbia University, Neural Acoustic Processing Laboratory, USA <i>Project: Neural Tracking of Linguistic Context in Attended and Unattended Multi-Talker Speech using Large Language Models</i> Supervisor: Prof. Nima Mesgarani
15/02/2021 - 25/06/2025	Doctoral researcher , Dept. of Neuroscience & Electrical Engineering, KU Leuven, ExpORL/PSI, Belgium Supervisors: Prof. Tom Francart and Prof. Hugo Van hamme
01/09/2019 – 15/03/2020	Master Thesis Researcher , Brown University, Brown Neuromotion Laboratory, USA Supervisor: Prof. David Borton
01/02/2019 – 15/06/2019	Student intern , EPFL, Center for Neuroprosthetics, Switzerland Supervisor: Prof. Grégoire Courtine

REFEREES

Prof. Tom Francart	tom.francart@kuleuven.be
Prof. Hugo Van hamme	hugo.vanhamme@kuleuven.be
Prof. Laura Gwilliams	laura.gwilliams@stanford.edu
Prof. Nima Mesgarani	nima@ee.columbia.edu

TEACHING - SUPERVISION

2026- *Sofie Sels, Master Student, KU Leuven*

SERVICE

Ad-hoc reviewer

Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT); Journal of NeuroEngineering and Rehabilitation; IEEE Transactions on Neural Systems and Rehabilitation Engineering (TNSRE)

RESEARCH OUTPUT

PEER-REVIEWED PUBLICATIONS

1. **C. Puffay**, G. Mischler, V. Choudhari, J. Vanthornhout, S. Bickel, A. D. Mehta, C. Schevon, G. M. McKhann, H. Van hamme, T. Francart, N. Mesgarani. Large Language Models Reveal the Neural Tracking of Linguistic Context in Attended and Unattended Multi-Talker Speech. *Imaging Neuroscience* (2026). Scientific Significance: Demonstrates how deep learning context representations track selective auditory attention in multi-talker settings, establishing a bridge between high-level generative AI context pipelines and human intracranial electrophysiology.
2. **C. Puffay**, J. Vanthornhout, M. Gillis, P. De Clercq, B. Accou, H. Van hamme, T. Francart. Classifying coherent versus nonsense speech perception from EEG using linguistic speech features. *Scientific Reports* 14, 18922 (2024). Scientific Significance: Introduced an analytical framework proving non-invasive EEG records can decode higher-level semantic comprehension, establishing a mechanism for objective cognitive hearing assessment without manual behavioral response.
3. **C. Puffay**, J. Vanthornhout, M. Gillis, B. Accou, H. Van hamme, T. Francart. Robust neural tracking of linguistic speech representations using a convolutional neural network. *Journal of Neural Engineering*, IOP Publishing (2023). Scientific Significance: Developed a highly robust convolutional model optimized for tracking rapid linguistic features from continuous brain recordings, substantially decreasing the recording duration required to map individualized speech-tracking metrics.
4. **C. Puffay**, B. Accou, L. Bollens, M.J. Monesi, J. Vanthornhout, H. Van hamme, T. Francart. Relating EEG to continuous speech using deep neural networks: a review. *Journal of Neural Engineering*, IOP Publishing (2023). Scientific Significance: Provided a foundational, highly cited paradigm synthesis of deep learning models linking natural continuous language structures to non-invasive electrophysiology.

CONFERENCE PROCEEDINGS

1. He, L., Nie, E., Dindar, S. S., Firoozi, A., Florea, A., Nguyen, V., **Puffay, C.**, Shimizu, R., Ye, H., Brennan, J., Schmid, H., Schütze, H., and Mesgarani, N. XCOMPS: A multilingual benchmark of conceptual minimal pairs. In *Proceedings of the 7th Workshop on Research in Computational Linguistic Typology and Multilingual NLP*, pages 75-81, Vienna, Austria (2025). Association for Computational Linguistics. Scientific Significance: Contributed to a multilingual language

processing benchmark designed to evaluate feature alignment against computational and neurobiological modeling frameworks.

2. P. De Clercq, **C. Puffay**, J. Kries, H. Van Hamme, M. Vandermosten, T. Francart, J. Vanthornhout. Detecting Post-Stroke Aphasia Via Brain Responses to Speech in a Deep Learning Framework. *IEEE Engineering in Medicine and Biology Conference (EMBC 2024)*. Scientific Significance: Developed an objective, deep-learning-based physiological screening pipeline to evaluate post-stroke language deficits using neural tracking metrics.
3. L. Bollens, **C. Puffay**, B. Accou, J. Vanthornhout, H. Van Hamme, T. Francart. Auditory EEG decoding challenge for ICASSP 2024. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2024)*. Scientific Significance: Co-conceptualized and managed a global baseline open challenge to benchmark competing machine learning architectures for decoding continuous speech from brain signals.
4. **C. Puffay**, J. Van Canneyt, J. Vanthornhout, H. Van hamme, T. Francart. Relating the fundamental frequency of speech using a dilated convolutional network. *Proc. Interspeech 2022*. Scientific Significance: Engineered an efficient, dilated convolutional neural network architecture tailored to capture low-level acoustic pitch tracking properties from early-stage auditory cortex EEG responses.

INVITED TALKS, CONFERENCE TALKS AND POSTERS AT ACADEMIC CONFERENCES

1. **Poster**. Single neuron encoding of acoustic and linguistic information in the human hippocampus during story listening. *Society for the Neurobiology of Language 18th annual meeting* (2026). Geneva, Switzerland. 29/09/2026-02/10/2026.
2. **Poster**. Large Language Models Embeddings as Predictors of Electrographic Responses in a Multi-Talker Attention Paradigm. *8th International Conference on Auditory Cortex 2025*. Maastricht, The Netherlands, 14/09/2025-18/09/2025.
3. **Invited Talk**. Diagnostics of the auditory system using deep-learning-based analyses of EEG signals. *Parra Lab, City College Of New York (CCNY), New York, USA*, 25/06/2024.
4. **Poster**. Neural tracking of linguistic features at different speech rates using a deep neural network. *Society for Neuroscience Annual Meeting (SfN)*, Washington DC, USA, 11/11/2023-15/11/2023.
5. **Poster**. Neural tracking of linguistic features at different speech rates using a deep neural network. *Audiological Research Cores in Europe (ARCHES) annual meeting*, Salamanca, Spain, 30/10/2023-31/10/2023.
6. **Poster**. Relating linguistic features of speech with EEG using a dilated convolutional network. *Society for Neuroscience Annual Meeting (SfN)*, San Diego, USA, 12/11/2022-16/11/2022.
7. **Conference Talk**. Relating the fundamental frequency of speech with EEG using a dilated convolutional network. *Interspeech 2022*. Incheon, South-Korea September 18-22th.

GRANTS AND AWARDS

27/07/2026

Postdoctoral Merit Award. Society for the Neurobiology of Language (SNL) 2026, 18th Annual Meeting, September 29 - October 2, Geneva, Switzerland.

14/03/2026 - 15/07/2026	Travel Grant for Long Stay Abroad. Research Foundation Flanders (FWO). <i>Funding to support an academic research stay at Stanford University (USA).</i>
2025	First Prize, EEG Foundation Challenge, NeurIPS 2025, San Diego, USA <i>First of >1,200 teams and >8,000 submissions. Team lead.</i>
14/06/2024 - 15/10/2024	Travel Grant for Long Stay Abroad. Research Foundation Flanders (FWO). <i>Funding to support an academic research stay at Columbia University (USA).</i>
01/10/2022 – Now	Personal Ph.D Fellowship for Strategic Basic Research. Research Foundation Flanders (FWO). Project: Diagnostics of the auditory system using deep-learning-based analysis of EEG signals.
2012	First Prize, French Physics Olympiads (Paris, France)